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Evaluation of the Balloon Analogue Risk Task (BART) as a predictor of adolescent real-world risk-taking behaviours

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Abstract

The Balloon Analogue Risk Task (BART) was administered to a sample of 26 high school-aged adolescents to test the utility of the BART as a behavioural measure of risk-taking propensity. Data indicate that riskyness on the BART was related to self-reported engagement in real-world risk-taking behaviours. These data replicate and uniquely extend past research to an adolescent sample, suggesting that the BART may be a useful addition to self-report batteries for the assessment of risk-taking behaviours in adolescents.
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1. Introduction

To date, the vast majority of research on adolescent risk-taking behaviour has relied on self-report methodologies to index risk-taking propensity (Jessor & Jessor, 1977; DiClemente, Hansen, & Ponton, 1995; Gullone & Moore, 2000). Despite the utility of self-report measures for these purposes there are a number of challenges to exclusively relying on self-reports of adolescents or their guardians, including response biases, inflation of method variance due to a unimodal assessment approach, strategic answering to appear less risky, and misunderstanding of questions. Currently, there are no behavioural measures that have been used with adolescents to identify risk-taking behaviours. To address these issues, we have developed the Balloon Analogue Risk Task (BART; Lejuez et al., 2002), a computer-based tool that indexes on-line responding by

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having subjects engage in simulated risk-taking behaviour. Previous research with young adults has shown that riskiness on the BART is positively correlated with scores on established risk-related constructs (e.g. sensation seeking, impulsivity) as well as risk-taking behaviours in the domains of substance use (e.g. smoking), delinquency (e.g. stealing), and safety (e.g. not wearing a seatbelt; Lejuez et al., 2002, 2003). In the present study, we extend this methodology to high-school aged adolescents and evaluated the degree to which responses on a battery of self-report measures and the BART were related to a composite of self-reported real-world risk-taking behaviours.

2. Method

2.1. Participants

Participants were 26 adolescents (15 females and 11 males) between the ages of 13 and 17 available at two after school programs. Of the participants, 13 (50%) were African American, 8 (31%) were primarily Caucasian, and the remaining 5 (19%) participants were either of Latino or Asian decent. Parental consent and adolescent assent were obtained for each participant.

2.2. Materials

Balloon Analogue Risk Task (BART; Lejuez et al., 2002):[1] The BART is presented as a computer-based measure of risk-taking propensity. Supporting its use in this regard, we have found in a sample of young adults between the ages of 18 and 25 that riskiness on the task is related to the self-reported engagement in a variety of real-world risk behaviours in the domains of substance use (e.g. smoking), delinquency (e.g. stealing), and safety (e.g. not wearing a seatbelt; Lejuez et al., 2002, 2003). During the task, the computer screen showed a small simulated balloon accompanied by a balloon pump, a reset button labeled "Collect $$$", a permanent money earned display labeled "Total Earned", and a second display listing the money earned on the last balloon and labeled "Last Balloon." With each pump, 5 cents were accrued in a temporary reserve (the amount of money in this reserve is never indicated to the participant). Once a balloon was pumped past its individual explosion point, a "pop" sound effect is emitted. When a balloon exploded, all money in the temporary bank was lost. At any point during each balloon, the participant could stop pumping the balloon and click the Collect $$$ button. Clicking this button transferred all money from the temporary bank to the permanent bank, during which the new total earned would be incrementally updated cent by cent while a slot machine payoff sound effect played. After each balloon explosion or money collection, the participant's exposure to that balloon ended and a new balloon appeared until all 30 balloons (i.e. trials) had been completed. All balloons had a different explosion point with the weakest balloon exploding on the first pump and the strongest balloon exploding after 128 pumps (see Lejuez et al., 2003, for further details). Participants were told that if they earned an amount of money above an undisclosed amount, that

[1] A computerized version of the Balloon Analogue Risk Task (BART) is available by request from the first author at no charge.

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they would receive 2 movie tickets. To prevent disappointment however, all participants were given the tickets after completing the task.

The primary dependent measure on the BART was the adjusted number of pumps across balloons (Lejuez et al., 2002). This adjusted value, defined as the average number of pumps excluding balloons that exploded (i.e. the average number of pumps on each balloon prior to collecting money), is preferable because the number of pumps is necessarily constrained on balloons that exploded, thereby limiting between-participant variability in the absolute averages. As a within session measure of test–retest reliability, previous research with the task has shown high correlations when the adjusted number of pumps is examined across the first, second, and third block of ten trials (average $r = 0.82$). In the current sample, the average correlation among the three blocks of trials was 0.86.

Self-report of risk-taking behaviours. The 10-item measure based on the Centers for Disease Control Youth Risk Behaviour Surveillance System (CDC, 2001), representing an index of engagement in risk-taking behaviours in daily life, was used to provide concurrent validation for the BART as a measure of risk-taking propensity. Specifically, participants answered yes or no to engagement in the following behaviours over the past 12 months: (1) smoked a cigarette (even a puff); (2) drank alcohol (even one drink); (3) used any illegal drug; (4) gambled for real money; (5) had sexual intercourse without a condom; (6) stolen anything from a store; (7) carried a weapon such as a gun, knife, or club outside of your home; (8) been in a physical fight; (9) ridden in a car without wearing your seatbelt (even once); (10) ridden a bicycle or motorcycle without a helmet (even once). A total risk behaviour index score was derived by summing the number of affirmative responses.

Self-report correlates of risk-taking behaviours. Assessing the construct validity of the BART, we used a battery of self-report measures aimed at tapping different constructs relevant to risk taking. To assess the seeking of varied, novel, complex, and intense situations and experiences, and the willingness to take physical, social, and financial risks for the sake of such experiences (Zuckerman, 1994), participants completed the *Sensation Seeking Scale* (Zuckerman, Eysenck, & Eysenck, 1978). On this 40-item measure, participants are given a forced choice between two opposite statements, with one choice indicative of sensation seeking. The number of times the sensation seeking choice is selected is summed to create a total score ranging from 0 to 40, with higher scores indicating higher levels of sensation seeking. The measure has good internal consistency with alpha coefficients above 0.8 for both males and females (Zuckerman et al., 1978; Zuckerman, 1994). To measure a propensity to engage in impulsive behaviour, participants completed the impulsivity subscale of the *Eysenck Impulsiveness Scale* (Eysenck, Pearson, Easting, & Allsopp, 1985). On this 19-item subscale, participants affirm or deny a particular impulsivity-related statement. Affirmative responses are summed to create a total subscale score ranging from 0 to 19, with higher scores indicating higher levels of impulsivity. This subscale has good internal consistency with an alpha coefficient equaling 0.84 (Eysenck et al., 1985). To measure global feelings of self-worth or self-regard, participants completed the *Rosenberg Self-Esteem Scale* (Rosenberg, 1965). This 10-item measure is scored on a 4-point likert scale with options ranging from strongly agree through strongly disagree. The total score on the measure ranges from 10 to 40, with higher scores indicating higher self-esteem. This measure is commonly used and adequate psychometric properties have been established across numerous studies (Gray-Little, Williams, & Hancock, 1997; Blascovich & Tomaka, 1991).

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Table 1
Means, standard deviations, and intercorrelations among each of the independent and dependent variables

	M (S.D.)	1	2	3	4	5
BART	33.0 (14.7)	—	0.48*	0.21	−0.12	0.33
RB	3.3 (2.2)		—	0.36	−0.15	0.56*
E-Imp	9.4 (3.7)			—	0.15	0.55*
SE	27.5 (7.7)				—	−0.33
SSS	17.3 (6.0)					—

Note. BART indicates adjusted average number of pumps on the Balloon Analogue Risk Task. RB indicates number of risk behaviors; E-Imp indicates the Impulsivity subscale score on the Eysenck Impulsiveness Scale; SE indicates total score on the Rosenberg Self-Esteem Scale; SSS indicates total score on the Sensation Seeking Scale.
* Indicates $p < 0.01$.

2.3. Procedure

Participants completed all aspects of the experiment in a private room. To limit the influence of order effects, questionnaire presentation order was randomized, with the BART occurring at some randomly determined point during this questionnaire completion period. The experimenter administering the BART was blind to participant responses on the self-report battery and the experimenter left the room while the participant was completing the questionnaires and the BART.

3. Results

Means and standard deviations for each measure are shown in Table 1. BART score did not differ as a function of gender, nor did the score for any of the self-report predictor measures. However, males ($M = 4.46$; S.D.=*2.02*) engaged in a significantly greater number of self-reported real-world risk-taking behaviours than females [$M = 2.40$; S.D.=*1.88*; $t(24) = 2.06$, $p = 0.01$]. Age ($M = 15.23$; S.D.=*1.03*) was not related to score on any of the above measures.

Also shown in Table 1, the BART score was not related to the score on the other self-report measures, but the correlation did approach significance with the Sensation Seeking total score ($p = 0.10$). Both BART score and Sensation Seeking total score were correlated with self-reported risk-taking behaviours. A moderate correlation between the Impulsivity subscale score and self-reported risk-taking behaviours also was observed, however, the modest sample size prevented statistical significance ($p = 0.07$).

4. Discussion

In the current study, score on the BART was related to self-reported engagement in real-world risk-taking behaviours, suggesting its potential utility as a measure of risk-taking propensity. Self-reported engagement in real-world risk-taking behaviours also was related to sensation seeking and its relationship with impulsivity approached significance. In addition to replicating previous

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research regarding sensation seeking and impulsivity, these data extend research with the BART to an adolescent sample. Although the use of a relatively small, yet ethnically diverse, sample of adolescents may preclude making any definitive statements about which measures are most related to real-world risk behaviours or the generalization of the present findings to all adolescents, the current results provide the requisite empirical basis upon which those tests can be conducted. Further, the fact that the BART was strongly related to the number of real-world risk-taking behaviours but showed only non-significant trends with several of the other measures suggests that the BART may be tapping an overlapping, yet conceptually different aspect of risk taking than that provided with measures of self-esteem, impulsivity, and sensation seeking. Taken together, the current results suggest that behavioural tasks such as the BART may be a useful addition to self-report batteries for assessing engagement in real-world risk-taking behaviours, which ultimately may have implications for the identification of individuals most in need of intervention strategies.

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References

Blascovich, J., & Tomaka, J. (1991). Measures of self-esteem. In J. P. Robinson, P. R. Shaver, & L. S. Wrightsman (Eds.), *Measures of personality and social psychological attitudes*, Vol. 1 (pp. 115–160). San Diego, CA: Academic Press.

Centers for Disease Control (CDC). (2001). *Youth risk behavior surveillance system*. Atlanta, GA: Centers for Disease Control.

DiClemente, R. J., Hansen, W. B., & Ponton, L. E. (1995). *Handbook of adolescent health risk behavior*. New York: Plenum.

Eysenck, S. B. G., Pearson, P. R., Easting, G., & Allsopp, J. F. (1985). Age norms for impulsiveness, venturesomeness, and empathy in adults. *Personality and Individual Differences*, *6*, 613–619.

Gray-Little, B., Williams, V. S. L., & Hancock, T. D. (1997). An item response theory analysis of the Rosenberg Self-Esteem Scale. *Personality and Social Psychology Bulletin*, *23*, 443–451.

Gullone, E., & Moore, S. (2000). Adolescent risk-taking and the five-factor model of personality. *Journal of Adolescence*, *23*, 393–407.

Jessor, S. L., & Jessor, R. (1977). *Problem behavior and psychosocial development: A longitudinal study of youth*. New York: Academic Press.

Lejuez, C. W., Aklin, W. M., Jones, H. A., Richards, J. R., Strong, D. R., Kahler, C. W., & Read, J. P. (2003). The Balloon Analogue Risk Task differentiates smokers and nonsmokers. *Experimental and Clinical Psychopharmacology*, *11*, 26–33.

Lejuez, C. W., Read, J. P., Kahler, C. W., Richards, J. B., Ramsey, S. E., Stuart, G. L., Strong, D. R., & Brown, R. A. (2002). Evaluation of a behavioral measure of risk-taking: The Balloon Analogue Risk Task (BART). *Journal of Experimental Psychology: Applied*, *6*, 75–84.

Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton, NJ: Princeton University Press.

Zuckerman, M. (1994). *Behavioral expressions and biosocial bases of sensation seeking*. New York: Cambridge University Press.

Zuckerman, M., Eysenck, S., & Eysenck, H. J. (1978). Sensation seeking in England and America: Cross-cultural, age, and sex comparisons. *Journal of Consulting and Clinical Psychology*, *46*, 139–149.

10959254, 2003, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1016/S0140-1971(03)00036-8 by National Dong Hwa University, Wiley Online Library on [22/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License